NAMAN YESHWANTH KUMAR

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SUMMARY

MS Computer Engineering (ASU, GPA 3.81, May 2026) at the intersection of autonomous vehicles, ML inference, and the silicon underneath. Implemented the full self-driving pipeline in C++ from scratch — sensor fusion (EKF), LiDAR localization (ICP, **0.299 m pose error / 4× baseline**), behavior FSM, and PID control — on the Waymo Open Dataset and CARLA. Hand-wrote 6 CUDA kernels for MLP training (manual cudaMalloc/cudaMemcpy + ctypes FFI) and built a 21-tool LangGraph agentic specialist for chip design that scored **37/92 on CVDP**.

TECHNICAL SKILLS

ML Systems & Inference: PyTorch | TensorFlow | CUDA (hand-written kernels) | nvcc | Python ↔ C++ FFI (ctypes) | NumPy | YOLOv4 | SSD ResNet50 | LangGraph | Multi-provider LLM Routing | MCP Protocol

Autonomous Vehicle Stack: C++ | LiDAR Point Cloud Processing | Bird's-Eye View | 3D Object Detection | Extended Kalman Filter | Multi-Object Tracking | ICP / SVD (from scratch) | NDT | Behavior FSM | Cubic Spiral Trajectories | PID Control | CARLA | Waymo Open Dataset | MuJoCo

Systems & Tools: C/C++ | Python | Linux | Docker | CMake | Eigen | PCL | GitHub Actions CI/CD | FastAPI | Coding Agents (Claude Code) | Git

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0) — **Expected May 2026**
Arizona State University — **Tempe, AZ**
Coursework: Perception in Robotics · GPU Architecture (CUDA) · Advanced Computer Architecture · Reinforcement Learning in Robotics · Algorithm/Hardware Co-Design

Bachelor of Engineering — Electrical and Electronics — **August 2022**
BMS College of Engineering — **Bengaluru, India**

PROJECT EXPERIENCE

Self-Driving Car Engineering Pipeline — C++, Waymo Open Dataset, CARLA Simulator — **Feb 2023**

- **Perception & Fusion:** Detected vehicles in 3D LiDAR point clouds via bird's-eye view projection using YOLOv4 (Darknet) and FPN ResNet18 on the Waymo Open Dataset; implemented an Extended Kalman Filter from scratch for LiDAR/camera fusion — full predict/update cycle, Mahalanobis data association, and track lifecycle (init/confirm/delete) validated via RMSE across multi-frame sequences.
- **Localization:** Implemented ICP from scratch using SVD decomposition (KdTree nearest-neighbor association, centroid alignment, iterative convergence) and benchmarked against NDT and PCL-based ICP; achieved **0.299 m pose error over 172 m traveled** (requirement <1.2 m) — **4× better than baseline**, earning "outstanding" reviewer feedback.
- **Planning & Control:** Built an FSM behavior planner (lane follow, nudge, lane change, intersection stop) with a motion planner generating cubic spiral trajectories evaluated by collision and lane-deviation cost functions; implemented PID controllers from scratch for steering (cross-track error) and throttle in CARLA.

CUDA MLP — Hand-Written GPU Kernels for Neural Network Training — **Fall 2024**

- Implemented 6 CUDA kernels in C++ by hand for end-to-end MLP training on MNIST — linear forward (matmul + bias), linear backward (gradients w.r.t. inputs/weights/biases), and ReLU forward/backward; configured block/thread launch dimensions per kernel, managed host↔device transfers explicitly with manual cudaMalloc/cudaMemcpy, and bound the kernels to a Python training loop through ctypes FFI — closing the loop from raw kernel code to a converged model.

SiliconCrew — Agentic Specialist Tooling for Hardware Design — **Nov 2025 – Present**

- Built a LangGraph ReAct agent (21 tools) that drives a complete RTL-to-GDSII flow with multi-provider LLM routing (Gemini/OpenAI/Anthropic), autonomous self-correction (read VCD → diagnose → patch → re-verify), and a full MCP server (stdio/SSE/HTTP) compatible with Claude Desktop, VS Code, and any MCP client; achieved **37/92 first-pass on CVDP** and **5/5 on ASU Spec2Tapeout**.

Graph Neural Network ASIC Accelerator — Full RTL-to-GDSII on ASAP7 7nm — **Spring 2025**

- Designed an 11-module GNN ASIC accelerator in SystemVerilog (8 parallel MACs + 12-state FSM for COO edge aggregation) and drove the full sign-off flow: Synopsys Design Compiler synthesis on ASAP7 7nm RVT, Cadence Innovus place-and-route, Voltus power analysis from VCD activity — **50,300 µm²** area, **52.29 mW**, **99.2 ns** latency.

Trezzit — Production AI Deployment with Per-Provider Cost & Latency Telemetry (trezzit.com) — **Feb 2025 – Present**

- Sole-built and shipped a production AI platform serving **600+ users** for 12+ months — multi-provider inference pipeline (Gemini, OpenAI, Claude, Grok) with two-pass semantic validation and automatic model-upgrade fallback (~60% → 95%+ accuracy); **995 automated tests** on GitHub Actions, per-endpoint p95 latency tracking, per-provider token/cost telemetry — full deployment + observability + rollout discipline at small scale.

WORK EXPERIENCE

Systems Engineer — **Oct 2022 – Apr 2024**
NCR GOLD — **Bengaluru, India**

- Automated the entire order lifecycle for a wholesale business — production Django platform (PostgreSQL + Celery + Nginx) processing **18,000+ orders over 3 years**; reverse-engineered third-party ERP (zero public API) via SQL schema analysis for bidirectional sync, eliminating ~20 min of manual relay per order.

AWARDS & HONORS

- Ranked **Top 10 nationally** in e-Yantra Robotics Competition (IIT Bombay) — designed a self-balancing vehicle in CoppeliaSim, February 2022.
- Won **'Best Use of AI'** at Strategy X DevHacks'25 (ASU) — built AdaptED AI, a Gemini-powered platform generating personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Won **1st place at Devlabs Hackathon** — HEAL.AI, a healthcare AI assistant with a custom RAG pipeline (768-dim embeddings, cosine similarity retrieval) for insurance Q&A and medical bill error detection.